

Database Schema - IT Help Desk & Ticketing Management System

Revision applied from instructor feedback: every table uses **Id** as its primary key. Foreign keys keep descriptive names so each relationship remains clear.

Database Tables

Table	Purpose	Primary Key	Important Columns
Role	Stores user roles.	Id	RoleName
UserAccount	Stores all users who can access the system.	Id	FullName, Email, PasswordHash, RoleId, Department, IsActive, CreatedDate
TicketCategory	Stores ticket category options.	Id	CategoryName
TicketPriority	Stores ticket urgency levels.	Id	PriorityName
TicketStatus	Stores ticket workflow states.	Id	StatusName
Ticket	Stores the main help desk support request.	Id	TicketNumber, Title, Description, CreatedByUserId, AssignedToUserId, TicketCategoryId, TicketPriorityId, TicketStatusId, CreatedDate, UpdatedDate, ResolvedDate, ClosedDate
TicketComment	Stores communication inside each ticket.	Id	TicketId, UserId, CommentText, IsInternal, CreatedDate
TicketAttachment	Stores files uploaded to tickets.	Id	TicketId, UploadedByUserId, FileName, FilePath, FileType, FileSizeBytes, UploadedDate
Notification	Stores user notifications.	Id	UserId, Title, Message, IsRead, CreatedDate
ActivityLog	Stores audit trail actions.	Id	UserId, TicketId, ActionName, ActionDetails, CreatedDate

Relationships

Parent	Child	Primary Key	Foreign Key	Relationship
Role	UserAccount	Role.Id	UserAccount.RoleId	One role can belong to many users.
UserAccount	Ticket	UserAccount.Id	Ticket.CreatedByUserId	One user can create many tickets.
UserAccount	Ticket	UserAccount.Id	Ticket.AssignedToUserId	One agent can be assigned to many tickets.
TicketCategory	Ticket	TicketCategory.Id	Ticket.TicketCategoryId	One category can classify many tickets.
TicketPriority	Ticket	TicketPriority.Id	Ticket.TicketPriorityId	One priority can be used by many tickets.
TicketStatus	Ticket	TicketStatus.Id	Ticket.TicketStatusId	One status can be used by many tickets.
Ticket	TicketComment	Ticket.Id	TicketComment.TicketId	One ticket can contain many comments.
Ticket	TicketAttachment	Ticket.Id	TicketAttachment.TicketId	One ticket can contain many attachments.

Parent	Child	Primary Key	Foreign Key	Relationship
UserAccount	Notification	UserAccount.Id	Notification.UserAccountId	One user can receive many notifications.
Ticket	ActivityLog	Ticket.Id	ActivityLog.TicketId	One ticket can have many activity logs.

SQL Server Schema Script

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CREATE TABLE Role (
    Id INT IDENTITY(1,1) NOT NULL,
    RoleName NVARCHAR(50) NOT NULL,
    CONSTRAINT PkRole_Id PRIMARY KEY (Id),
    CONSTRAINT UnRole_RoleName UNIQUE (RoleName)
);

CREATE TABLE UserAccount (
    Id INT IDENTITY(1,1) NOT NULL,
    FullName NVARCHAR(100) NOT NULL,
    Email NVARCHAR(150) NOT NULL,
    PasswordHash NVARCHAR(MAX) NOT NULL,
    RoleId INT NOT NULL,
    Department NVARCHAR(100) NULL,
    IsActive BIT NOT NULL CONSTRAINT DfUserAccount_IsActive DEFAULT 1,
    CreatedDate DATETIME2 NOT NULL
    CONSTRAINT DfUserAccount_CreatedDate DEFAULT SYSUTCDATETIME(),
    CONSTRAINT PkUserAccount_Id PRIMARY KEY (Id),
    CONSTRAINT UnUserAccount_Email UNIQUE (Email),
    CONSTRAINT FkUserAccount_RoleId
        FOREIGN KEY (RoleId) REFERENCES Role(Id)
);

CREATE TABLE TicketCategory (
    Id INT IDENTITY(1,1) NOT NULL,
    CategoryName NVARCHAR(100) NOT NULL,
    CONSTRAINT PkTicketCategory_Id PRIMARY KEY (Id),
    CONSTRAINT UnTicketCategory_CategoryName UNIQUE (CategoryName)
);

CREATE TABLE TicketPriority (
    Id INT IDENTITY(1,1) NOT NULL,
    PriorityName NVARCHAR(50) NOT NULL,
    CONSTRAINT PkTicketPriority_Id PRIMARY KEY (Id),
    CONSTRAINT UnTicketPriority_PriorityName UNIQUE (PriorityName)
);

CREATE TABLE TicketStatus (
    Id INT IDENTITY(1,1) NOT NULL,
    StatusName NVARCHAR(50) NOT NULL,
    CONSTRAINT PkTicketStatus_Id PRIMARY KEY (Id),
    CONSTRAINT UnTicketStatus_StatusName UNIQUE (StatusName)
);

CREATE TABLE Ticket (
    Id INT IDENTITY(1,1) NOT NULL,
    TicketNumber NVARCHAR(20) NOT NULL,
    Title NVARCHAR(200) NOT NULL,
    Description NVARCHAR(MAX) NOT NULL,
    CreatedByUserAccountId INT NOT NULL,
    AssignedToUserAccountId INT NULL,
    TicketCategoryId INT NOT NULL,
    TicketPriorityId INT NOT NULL,
    TicketStatusId INT NOT NULL,
    CreatedDate DATETIME2 NOT NULL
    CONSTRAINT DfTicket_CreatedDate DEFAULT SYSUTCDATETIME(),
    UpdatedDate DATETIME2 NOT NULL
    CONSTRAINT DfTicket_UpdatedDate DEFAULT SYSUTCDATETIME(),
    ResolvedDate DATETIME2 NULL,
    ClosedDate DATETIME2 NULL,
    CONSTRAINT PkTicket_Id PRIMARY KEY (Id),
    CONSTRAINT UnTicket_TicketNumber UNIQUE (TicketNumber),
    CONSTRAINT FkTicket_CreatedByUserAccountId
        FOREIGN KEY (CreatedByUserAccountId) REFERENCES UserAccount(Id),
    CONSTRAINT FkTicket_AssignedToUserAccountId
        FOREIGN KEY (AssignedToUserAccountId) REFERENCES UserAccount(Id),
    CONSTRAINT FkTicket_TicketCategoryId
        FOREIGN KEY (TicketCategoryId) REFERENCES TicketCategory(Id),
    CONSTRAINT FkTicket_TicketPriorityId
        FOREIGN KEY (TicketPriorityId) REFERENCES TicketPriority(Id),
    CONSTRAINT FkTicket_TicketStatusId
        FOREIGN KEY (TicketStatusId) REFERENCES TicketStatus(Id)
);

CREATE TABLE TicketComment (
    Id INT IDENTITY(1,1) NOT NULL,
    TicketId INT NOT NULL,
    UserAccountId INT NOT NULL,
    CommentText NVARCHAR(MAX) NOT NULL,
    IsInternal BIT NOT NULL CONSTRAINT DfTicketComment_IsInternal DEFAULT 0,
    CreatedDate DATETIME2 NOT NULL
    CONSTRAINT DfTicketComment_CreatedDate DEFAULT SYSUTCDATETIME(),
    CONSTRAINT PkTicketComment_Id PRIMARY KEY (Id),
    CONSTRAINT FkTicketComment_TicketId
        FOREIGN KEY (TicketId) REFERENCES Ticket(Id),
    CONSTRAINT FkTicketComment_UserAccountId
        FOREIGN KEY (UserAccountId) REFERENCES UserAccount(Id)
);

CREATE TABLE TicketAttachment (
    Id INT IDENTITY(1,1) NOT NULL,

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TicketId INT NOT NULL,
UploadedByUserId INT NOT NULL,
FileName NVARCHAR(255) NOT NULL,
FilePath NVARCHAR(500) NOT NULL,
FileType NVARCHAR(100) NOT NULL,
FileSizeBytes INT NOT NULL,
UploadDate DATETIME2 NOT NULL
CONSTRAINT DfTicketAttachment_UploadDate DEFAULT SYSUTCDATETIME(),
CONSTRAINT PkTicketAttachment_Id PRIMARY KEY (Id),
CONSTRAINT FkTicketAttachment_TicketId
    FOREIGN KEY (TicketId) REFERENCES Ticket(Id),
CONSTRAINT FkTicketAttachment_UploadedByUserId
    FOREIGN KEY (UploadedByUserId) REFERENCES UserAccount(Id),
CONSTRAINT CkTicketAttachment_FileSizeBytes CHECK (FileSizeBytes > 0)
);

CREATE TABLE Notification (
    Id INT IDENTITY(1,1) NOT NULL,
    UserId INT NOT NULL,
    Title NVARCHAR(200) NOT NULL,
    Message NVARCHAR(MAX) NOT NULL,
    IsRead BIT NOT NULL CONSTRAINT DfNotification_IsRead DEFAULT 0,
    CreatedDate DATETIME2 NOT NULL
CONSTRAINT DfNotification_CreatedDate DEFAULT SYSUTCDATETIME(),
CONSTRAINT PkNotification_Id PRIMARY KEY (Id),
CONSTRAINT FkNotification_UserId
    FOREIGN KEY (UserId) REFERENCES UserAccount(Id)
);

CREATE TABLE ActivityLog (
    Id INT IDENTITY(1,1) NOT NULL,
    UserId INT NOT NULL,
    TicketId INT NOT NULL,
    ActionName NVARCHAR(100) NOT NULL,
    ActionDetails NVARCHAR(MAX) NULL,
    CreatedDate DATETIME2 NOT NULL
CONSTRAINT DfActivityLog_CreatedDate DEFAULT SYSUTCDATETIME(),
CONSTRAINT PkActivityLog_Id PRIMARY KEY (Id),
CONSTRAINT FkActivityLog_UserId
    FOREIGN KEY (UserId) REFERENCES UserAccount(Id),
CONSTRAINT FkActivityLog_TicketId
    FOREIGN KEY (TicketId) REFERENCES Ticket(Id)
);
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